Chapter 17, Problem 49P

Problem

(a) For the periodic waveform in Prob. 17.5, find the rms value.

(b) Use the first five harmonic terms of the Fourier series in Prob. 17.5 to determine the effective value of the signal.

(c) Calculate the percentage error in the estimated rms value of *z*(*t*) if

$$\%\text{ error} = \left(\frac{\text{estimated value}}{\text{exact value}} - 1\right) \times 100$$

Step-by-step solution

Step 1 of 8

Refer to Figure 17.49 in the textbook for square wave.

The function is described as,

$$f(t) = \begin{cases} 7, & 0 < t < \pi \\ -14, & \pi < t < 2\pi \end{cases}$$

(a)

Calculate the rms value.

$$\begin{aligned} F_{\text{rms}} &= \sqrt{\frac{1}{T}\int_0^T f^2(t)\,dt} \\ &= \sqrt{\frac{1}{2\pi}\left[\int_0^{\pi} f^2(t)\,dt + \int_{\pi}^{2\pi} f^2(t)\,dt\right]} \\ &= \sqrt{\frac{1}{2\pi}\left[\int_0^{\pi} 7^2\,dt + \int_{\pi}^{2\pi} (-14)^2\,dt\right]} \\ &= \sqrt{\frac{1}{2\pi}\left[49\pi + 196\pi\right]} \end{aligned}$$

$= 11.068\text{ V}$

Therefore, the rms value, F_{rms} is $\boxed{11.068\text{ V}}$.

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(b)

Time period of the waveform is, $T = 2\pi$

Calculate the angular frequency.

$$\begin{aligned}\omega &= \frac{2\pi}{T}\\ &= \frac{2\pi}{2\pi}\\ &= 1 \text{ rad/s}\end{aligned}$$

Calculate the Fourier coefficient, a_0.

$$\begin{aligned}a_0 &= \frac{1}{2\pi}\int_0^T f(t)\,dt\\ &= \frac{1}{2\pi}\int_0^{\pi} f(t)\,dt + \frac{1}{2\pi}\int_{\pi}^{2\pi} f(t)\,dt\\ &= \frac{1}{2\pi}\int_0^{\pi} 7\,dt + \frac{1}{2\pi}\int_{\pi}^{2\pi} (-14)\,dt\end{aligned}$$

$$\begin{aligned}&= \frac{1}{2\pi}(7\pi - 14\pi)\\ &= -\frac{7}{2}\end{aligned}$$

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Calculate the Fourier coefficient, a_n.

$$\begin{aligned}a_n &= \frac{2}{2\pi}\int_0^T f(t)\cos(n\omega_0 t)\,dt\\ &= \frac{1}{\pi}\int_0^{2\pi} f(t)\cos(nt)\,dt\\ &= \frac{1}{\pi}\int_0^{\pi} 7\cos(nt)\,dt + \frac{1}{\pi}\int_{\pi}^{2\pi} (-14)\cos(nt)\,dt\end{aligned}$$

$$\begin{aligned}&= \frac{7}{n\pi}\left[\sin(nt)\right]_{t=0}^{t=\pi} - \frac{14}{n\pi}\left[\sin(nt)\right]_{t=\pi}^{t=2\pi}\\ &= 0\end{aligned}$$

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Calculate the Fourier coefficient, b_n.

$$\begin{aligned}b_n &= \frac{2}{2\pi}\int_0^T f(t)\cos(n\omega_0 t)\,dt\\ &= \frac{1}{\pi}\int_0^{2\pi} f(t)\sin(nt)\,dt\\ &= \frac{1}{\pi}\int_0^{\pi} 7\sin(nt)\,dt + \frac{1}{\pi}\int_{\pi}^{2\pi} (-14)\sin(nt)\,dt\end{aligned}$$

$$\begin{aligned}&= \frac{7}{n\pi}\left[-\cos(nt)\right]_{t=0}^{t=\pi} - \frac{14}{n\pi}\left[-\cos(nt)\right]_{t=\pi}^{t=2\pi}\\ &= \frac{7}{n\pi}(1-\cos(n\pi)) + \frac{14}{n\pi}(1-\cos(n\pi))\\ &= \frac{21}{n\pi}(1-\cos(n\pi))\\ &= \frac{42}{n\pi}, n \text{ is odd}\end{aligned}$$

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Therefore, the Fourier series is,

$$\begin{aligned} f(t) &= a_0 + \sum_{n=1}^{\infty}\left(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t\right) \\ &= -\frac{7}{2} + \sum_{\substack{n=1 \\ n=\text{odd}}}^{\infty}\left(b_n \sin nt\right) \\ &= -\frac{7}{2} + \sum_{\substack{n=1 \\ n=\text{odd}}}^{\infty}\left\{\frac{42}{n\pi}\sin(nt)\right\} \\ &= -\frac{7}{2} + \frac{42}{\pi}\sum_{\substack{n=1 \\ n=\text{odd}}}^{\infty}\left\{\frac{1}{n}\sin(nt)\right\} \end{aligned}$$

Calculate the rms value of first five harmonic terms.

$$\begin{aligned} F_{\text{rms}} &= \sqrt{a_0^2 + \sum_{n=1}^{\infty}\frac{1}{2}\left(a_n^2 + b_n^2\right)} \\ &= \sqrt{\left(-\frac{7}{2}\right)^2 + \sum_{n=1}^{\infty}\frac{1}{2}\left(0 + b_n^2\right)} \end{aligned}$$

$$F_{\text{rms}} = \sqrt{\left(-\frac{7}{2}\right)^2 + \frac{1}{2}\left(b_1^2 + b_2^2 + b_3^2 + b_4^2 + b_5^2\right)} \quad \ldots\ldots (1)$$

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Calculate the harmonic terms.

$$\begin{aligned} b_1 &= \frac{42}{\pi} \\ &= 13.369 \\ b_3 &= \frac{42}{3\pi} \\ &= 4.4563 \\ b_5 &= \frac{42}{5\pi} \\ &= 2.6738 \\ b_7 &= \frac{42}{7\pi} \\ &= 1.9099 \\ b_9 &= \frac{42}{9\pi} \\ &= 1.4854 \end{aligned}$$

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From equation (1),

$$\begin{aligned} F_{\text{rms}} &= \sqrt{3.5^2 + \frac{1}{2}\left(13.369^2 + 4.4563^2 + (2.6738)^2 + (1.9099)^2 + (1.4854)^2\right)} \\ &= \sqrt{3.5^2 + \frac{1}{2}(211.59)} \\ &= \sqrt{12.25 + 105.79} \\ &= 10.865 \end{aligned}$$

Therefore, the rms value of first five harmonic terms is $\boxed{10.865}$.

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(c)

The percentage of error in the estimated rms value of $f(t)$ is,

$$\begin{aligned}\%\,\text{error} &= \left(\frac{\text{estimated value}}{\text{exact value}} - 1\right)\times 100\\ &= \left(\frac{10.865}{11.068} - 1\right)\times 100\\ &= -1.835\%\end{aligned}$$

Therefore, the percentage of error in the estimated rms value is $\boxed{-1.835\%}$.